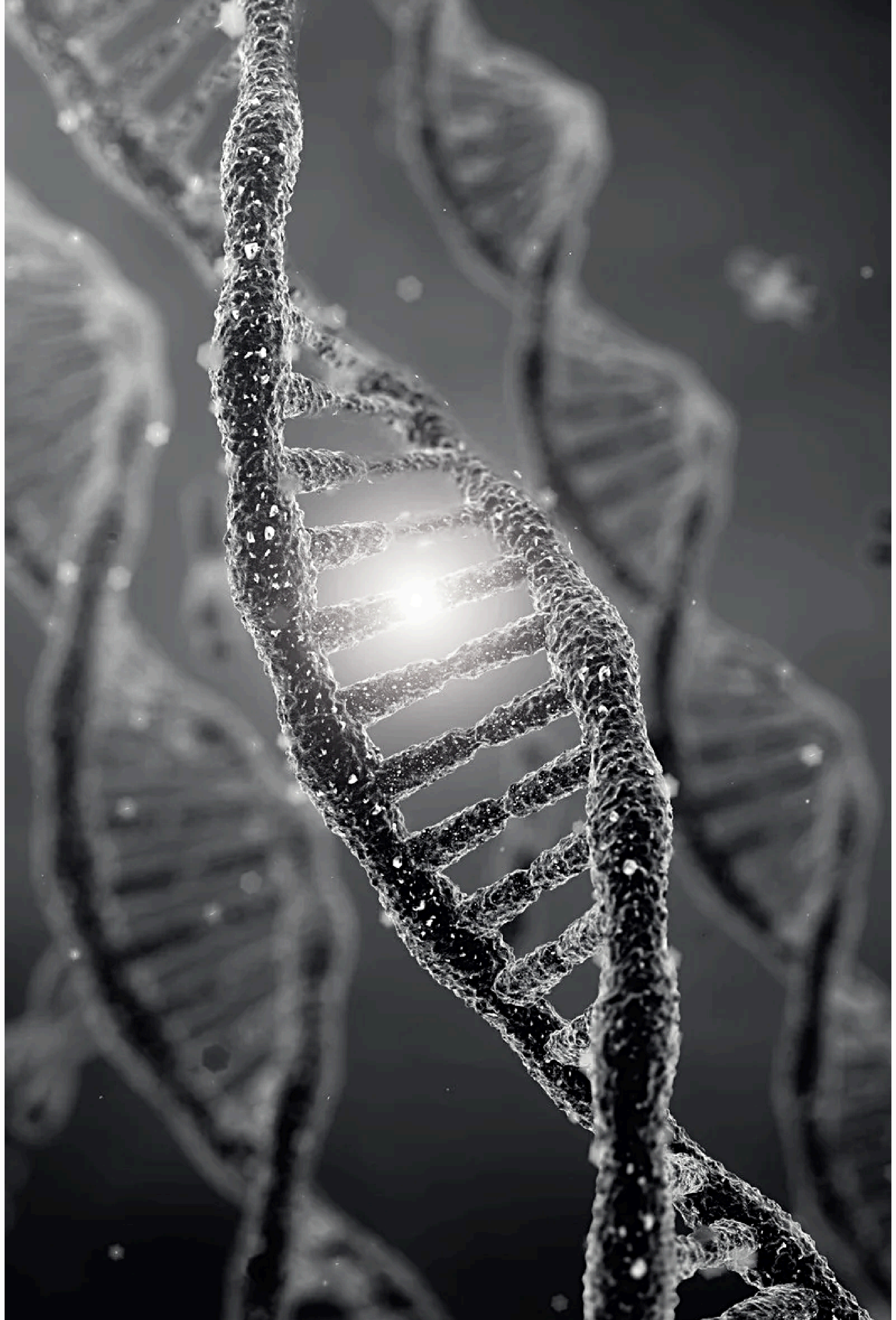


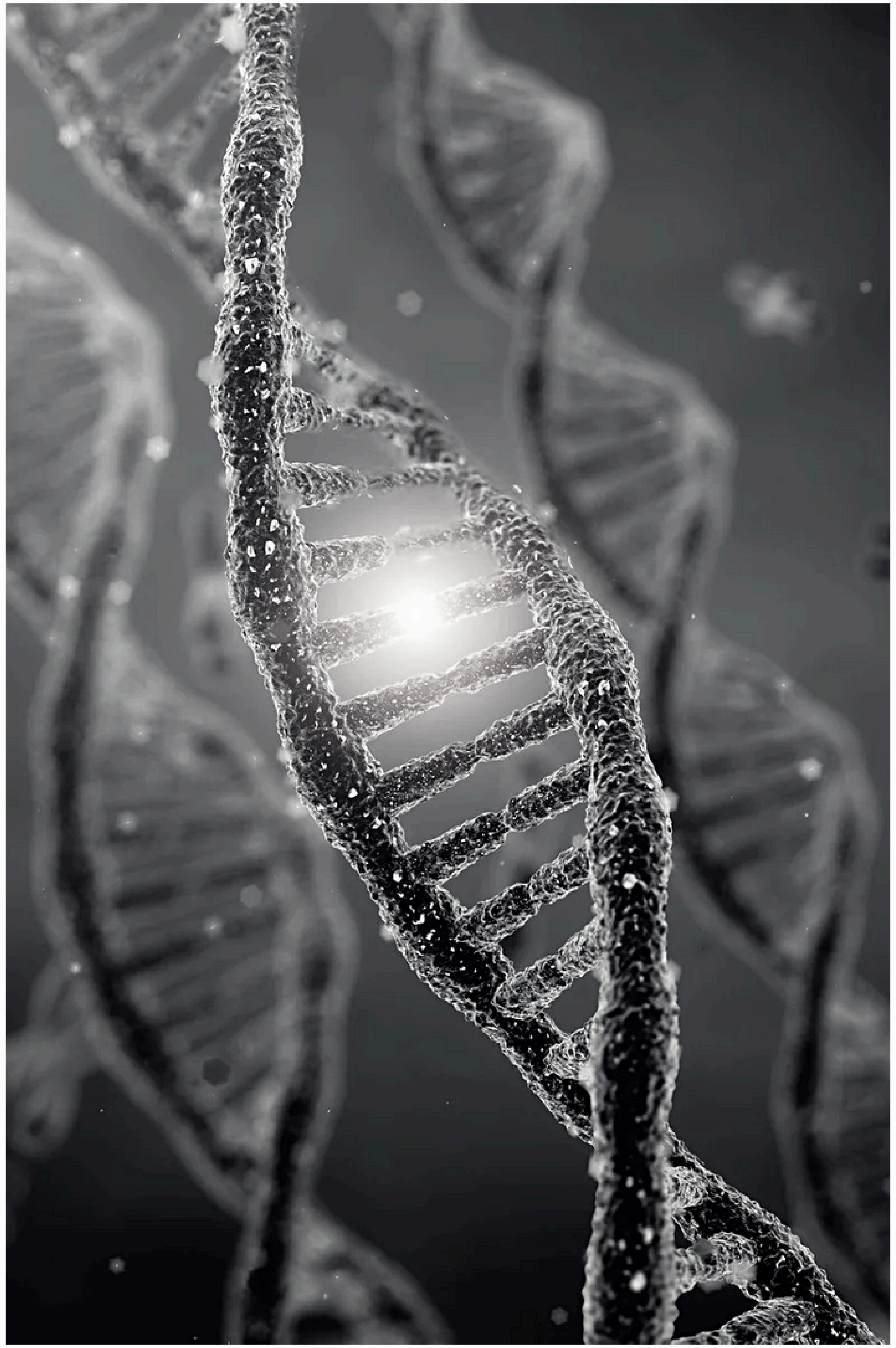
Epigenetic Clock

Alisia Sallemi 866453

Overview

List of key concepts

- What is epigentics?
- What is DNA methylation?
- What is an epigenetic clock?
- Workflow
- Data retrieval and management
- Epigenetic clock model
- Results analysis



What is **Epigenetics**?

Epigenetics is the study of the mechanisms that regulate **gene expression** without changing the DNA sequence.

Every cell in our body contains the same DNA, but epigenetics determines what genes are "turned on" or "turned off", and thus what function a cell will have (e.g., muscle, nerve, etc.).

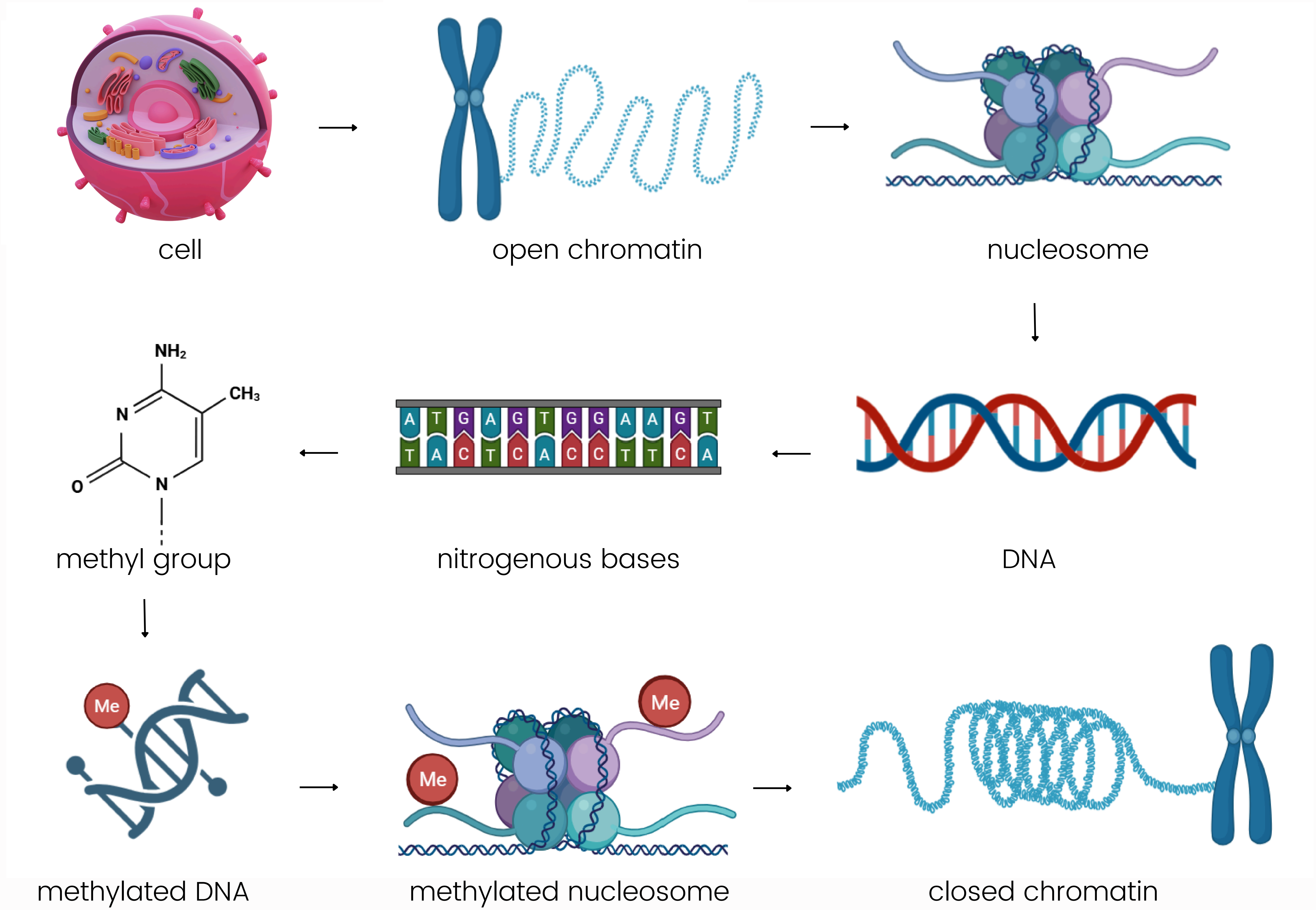
Can epigentic mutations be predictive of age?

What is DNA methylation?

DNA methylation is an epigenetic mechanism that involves the addition of a methyl group ($-CH_3$) to specific DNA sites, called CpG sequences (where a cytosine is followed by a guanine).

This process regulates gene expression by silencing genes, preventing transcription and protein synthesis without altering the DNA sequence itself.

The reverse process, **DNA demethylation**, promotes increased transcription and protein synthesis.


cell
open chromatin
nucleosome
NH2
CH3
N
O
N
methyl group
A T G A G T G G A A G T
T A C T C A C C T T C A
nitrogenous bases
DNA
Me
methylated DNA
Me
Me
methylated nucleosome
closed chromatin

What is an **epigenetic clock**?

Epigenetic clocks are statistical and computational tools that estimate the biological age of an individual based on **DNA methylation** patterns, particularly **CpG sites**.

Biological age may differ from chronological age indicative of accelerated or decelerated aging.

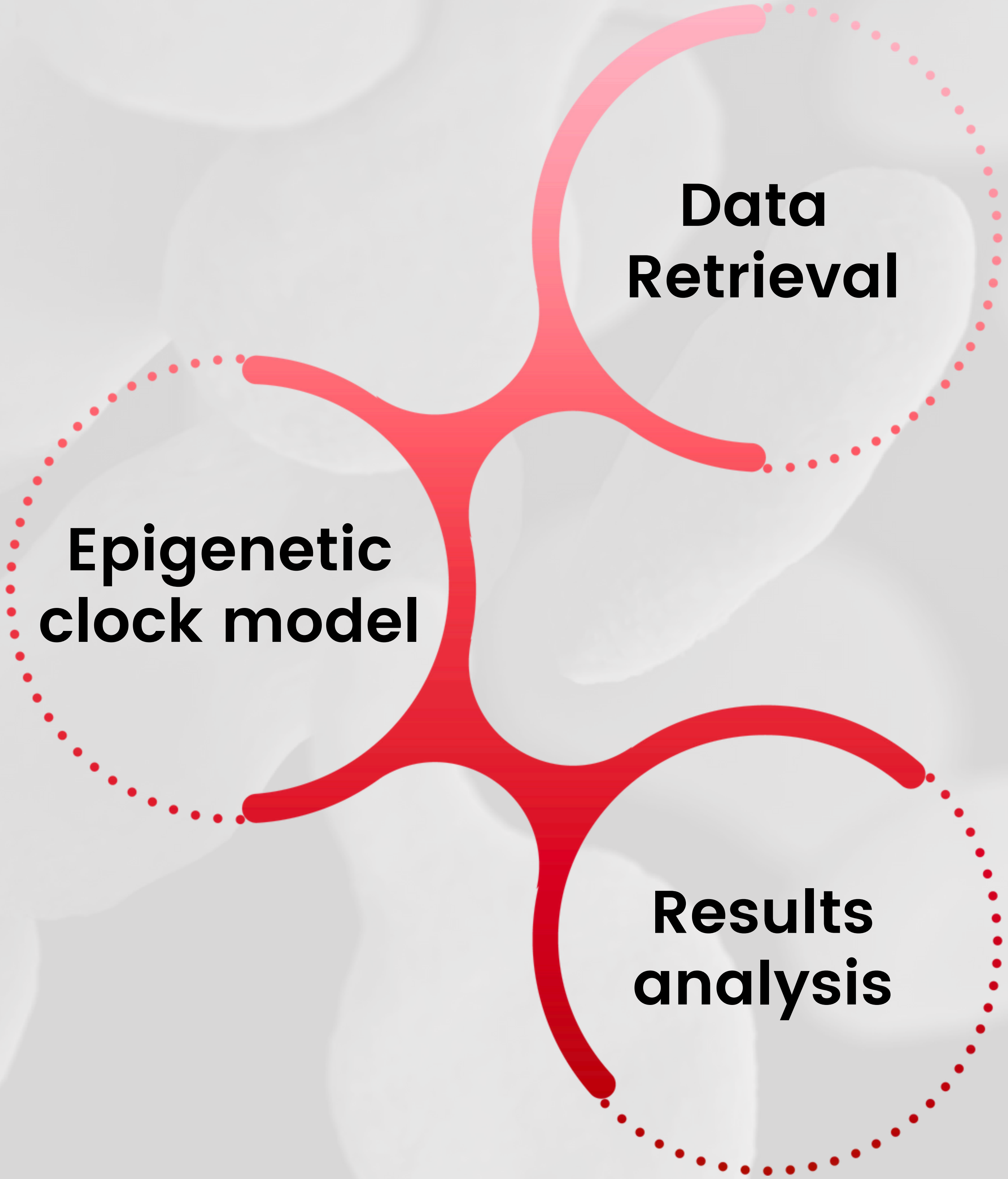
The first well-known epigenetic clock is that of **Horvath** (2013) who identified a set of **353 CpG sites** predictive of age through machine learning models.

Project Aim

Leverage DNA methylation changes to uncover the underlying drivers of aging.

Build a robust model that is adaptable to emerging epigenetic profiling technologies.

Workflow



Data Retrieval and Management

Data acquisition

Data were downloaded from NCBI GEO with the R package GEOquery specifying the GEO accession code: GSE246337 and GSE280465.

Data type

DNA methylation data are obtained through Infinium MethylationEPIC v2.0 array platform.

Metadata

Metadata were cleaned to keep only relevant features: geo accession, subject id, tissue, sex, race, ethnicity, sample id, chronological age.

Data**sets**

GSE246337
This dataset contains DNA methylation values derived from **blood** cells of **500 individuals.**

For each subject, information on chronological age, sex, race, and ethnicity is available.
The age range spans from **18 to 89 years**.

This dataset was used to construct the **training** and **validation** sets.

GSE280465
This dataset includes DNA methylation data obtained from **four different tissues**: saliva, blood, fingerstick blood, and buccal epithelial cells, collected from **47 individuals**, totaling 163 samples.

For each subject, information on chronological age and sex is available.
The age range spans from **19 to 70 years.**

Only the blood samples were selected from this dataset and used for the **test set.**

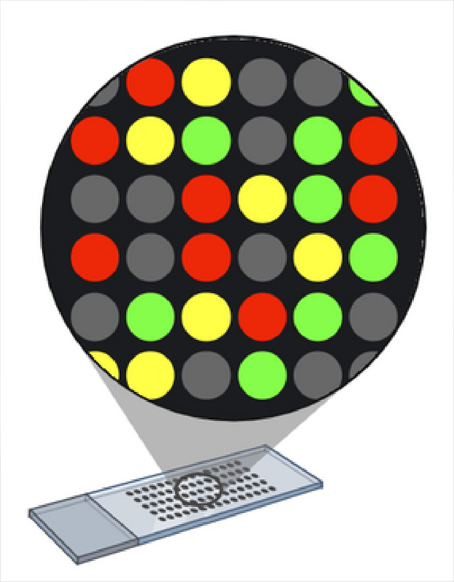
Data **Type**

Infinium MethylationEPIC v2.0 arrays are a high-density technology used to quantify the methylation of CpG sites throughout the human genome.
The EPIC v2.0 version improves coverage to over 935,000 CpG sites.

illumina

How do they work?

1. Genomic DNA first undergoes **bisulfite conversion**:
   a. **unmethylated** cytosines are converted to uracils,
   b. **methylated** cytosines remain unchanged.
2. The treated DNA is **hybrided** on the array, which contains specific probes for each CpG site.
3. Infinium probes function according to two types:
   a. **Type I**: two separate probes for methylated and unmethylated CpG.
   b. **Type II**: one probe with discrimination by fluorescence (red/green).
4. Methylation level is calculated as **β-value**, from 0 (unmethylated) to 1 (fully methylated).



Epigenetic Clock Model

Data normalization

β-values were computed by sample-wise normalization, using the sesame package.

Dataset division

The dataset of 500 samples was split into a training set (80%) and a validation set (20%).

The split was performed to ensure that:

- the mean and standard deviation of age were comparable between the two sets,
- sex was evenly distributed across training and validation sets.

Missing values imputation

Missing values, resulting from low measurement precision of the signal (high p-values), were imputed using the K-Nearest Neighbors (KNN) algorithm with the 5 most similar neighbors.

Epigenetic Clock Model

Model training and validation

The model used was an elastic net regression with α = 0.5, trained through 20-fold cross-validation.

Model testing and comparison

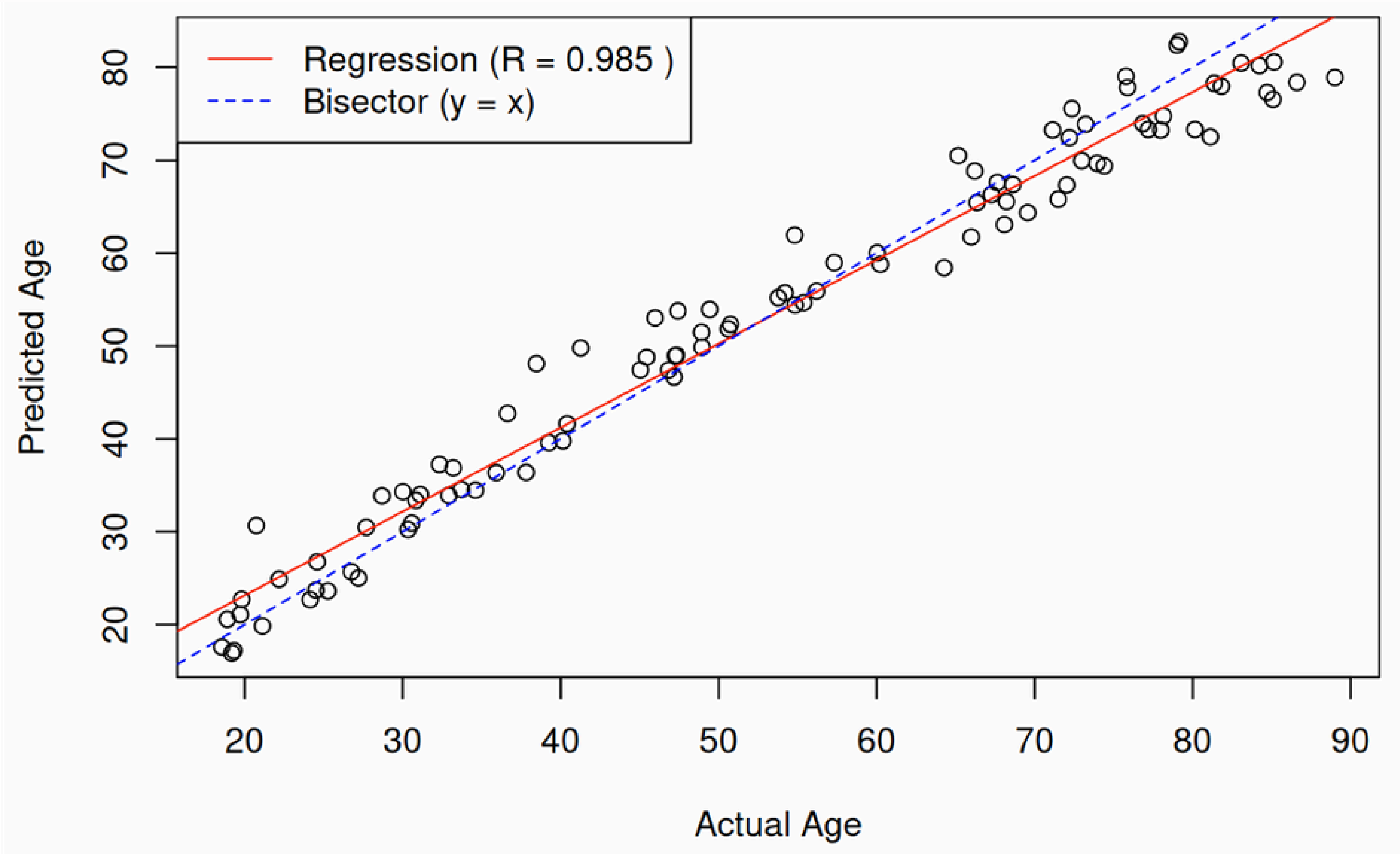
The model was tested on the second dataset, and the results were **compared** with predictions obtained on the same samples using two well-known epigenetic models developed by **Steve Horvath**:

- the first one is optimized for **pan-tissue** age prediction,
- while the second is specifically designed for data derived from **epithelial and blood cells.**

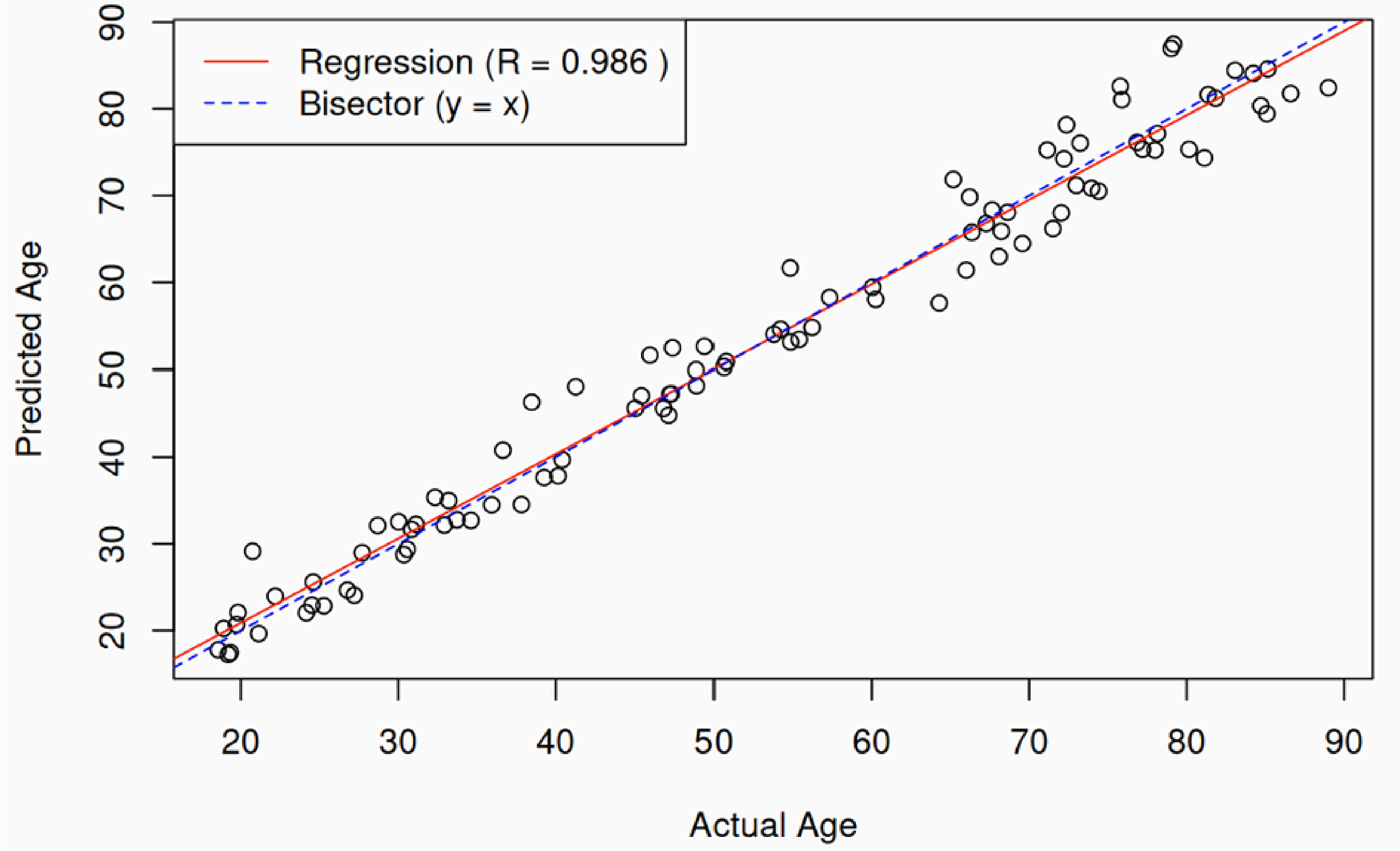
Model validation and Calibration

A linear and systematic bias in predictions (compression at extremes) was corrected via post-hoc calibration, using a polynomial regression.

The same calibration was applied to the test set aswell.



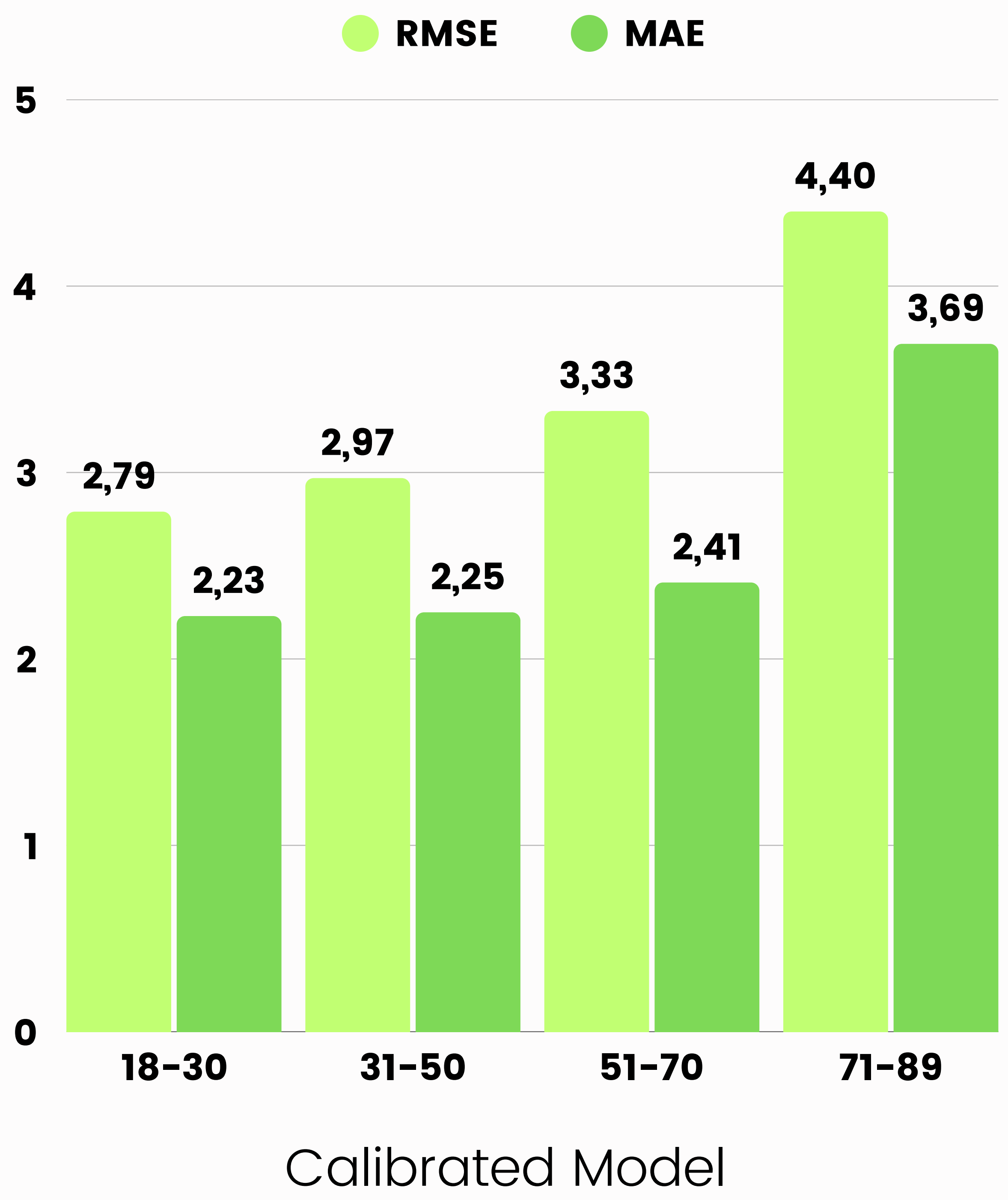
Before calibration



After calibration

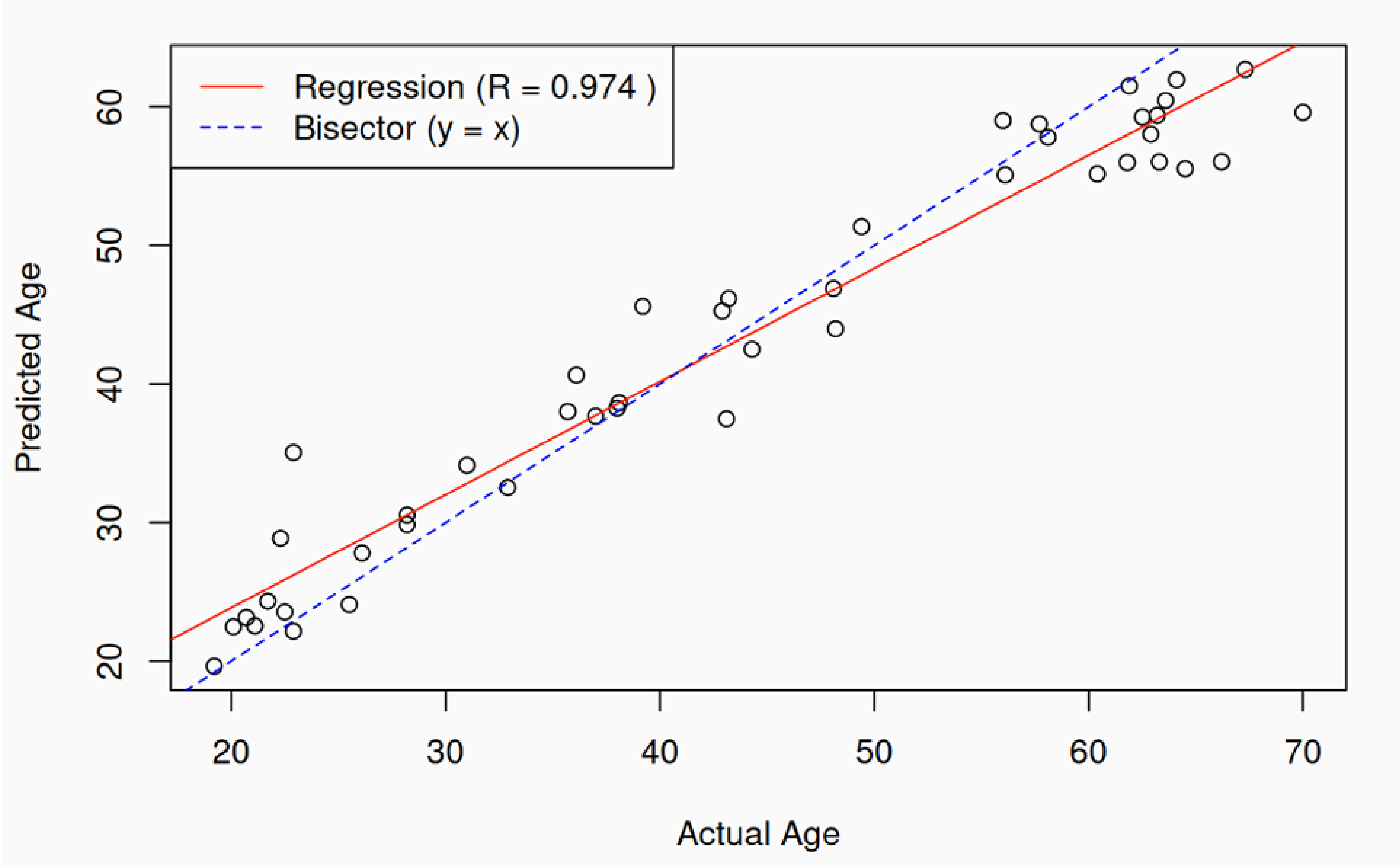
Overall results

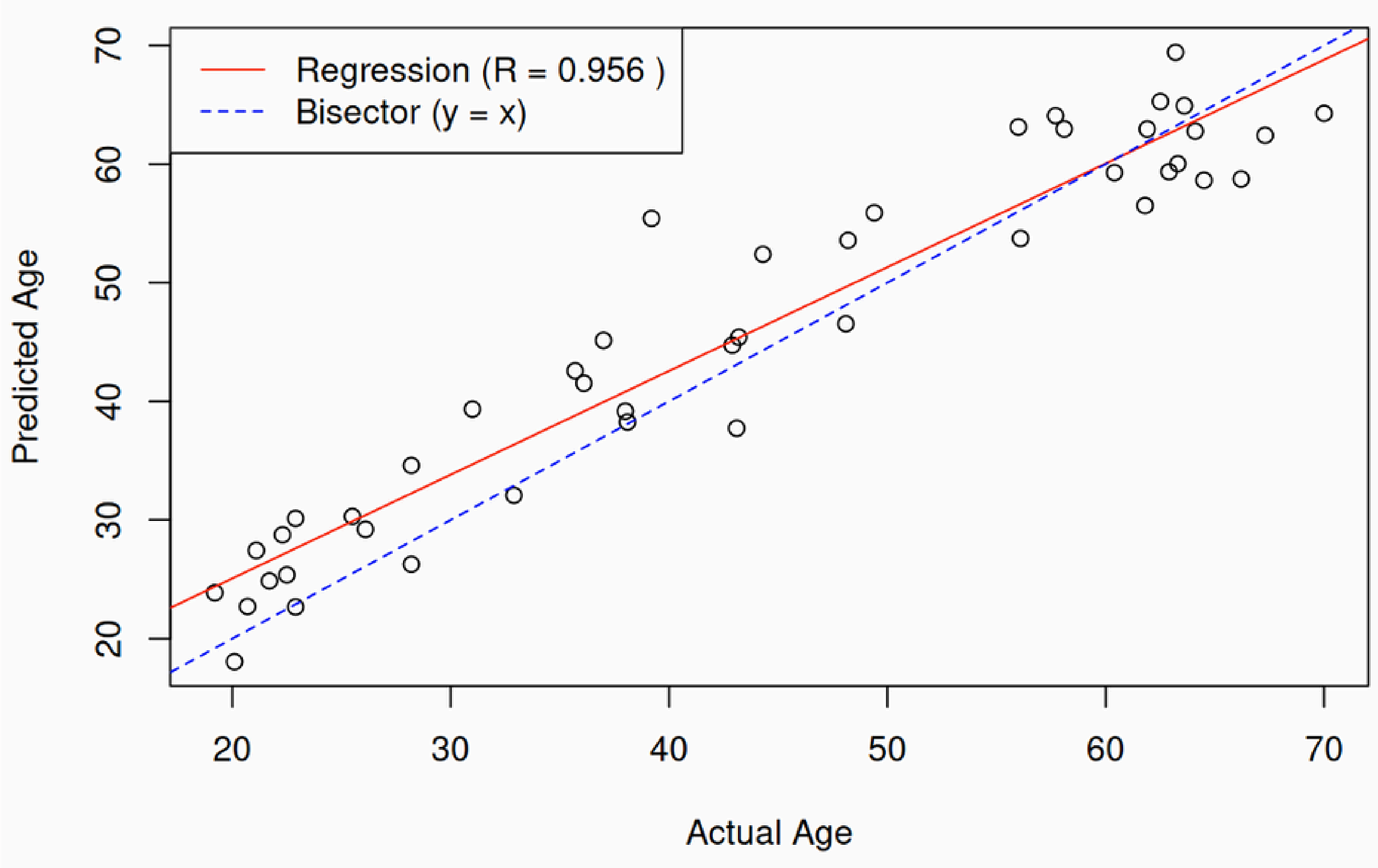
Results by age group



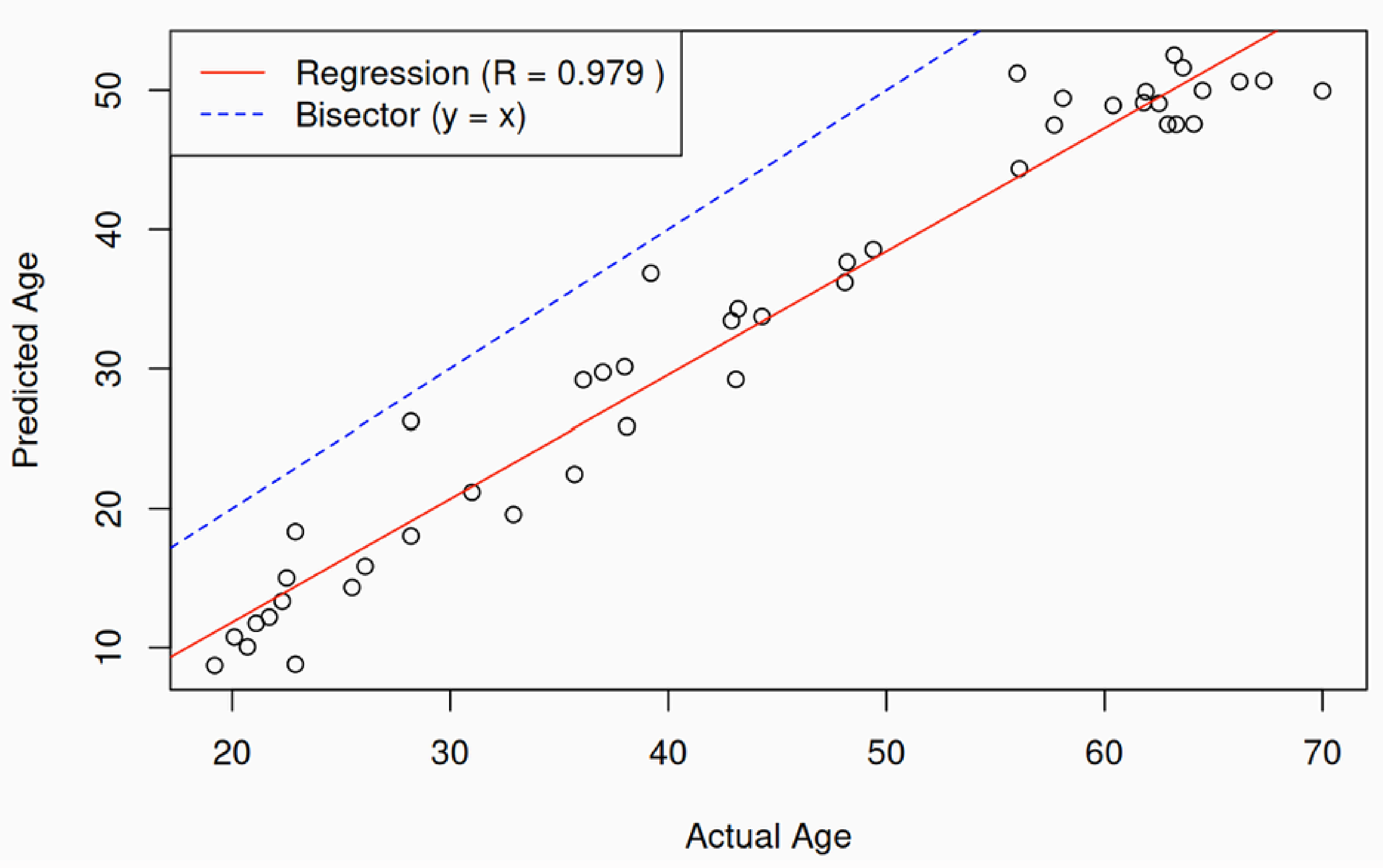
Model Testing and Comparison

Calibrated model



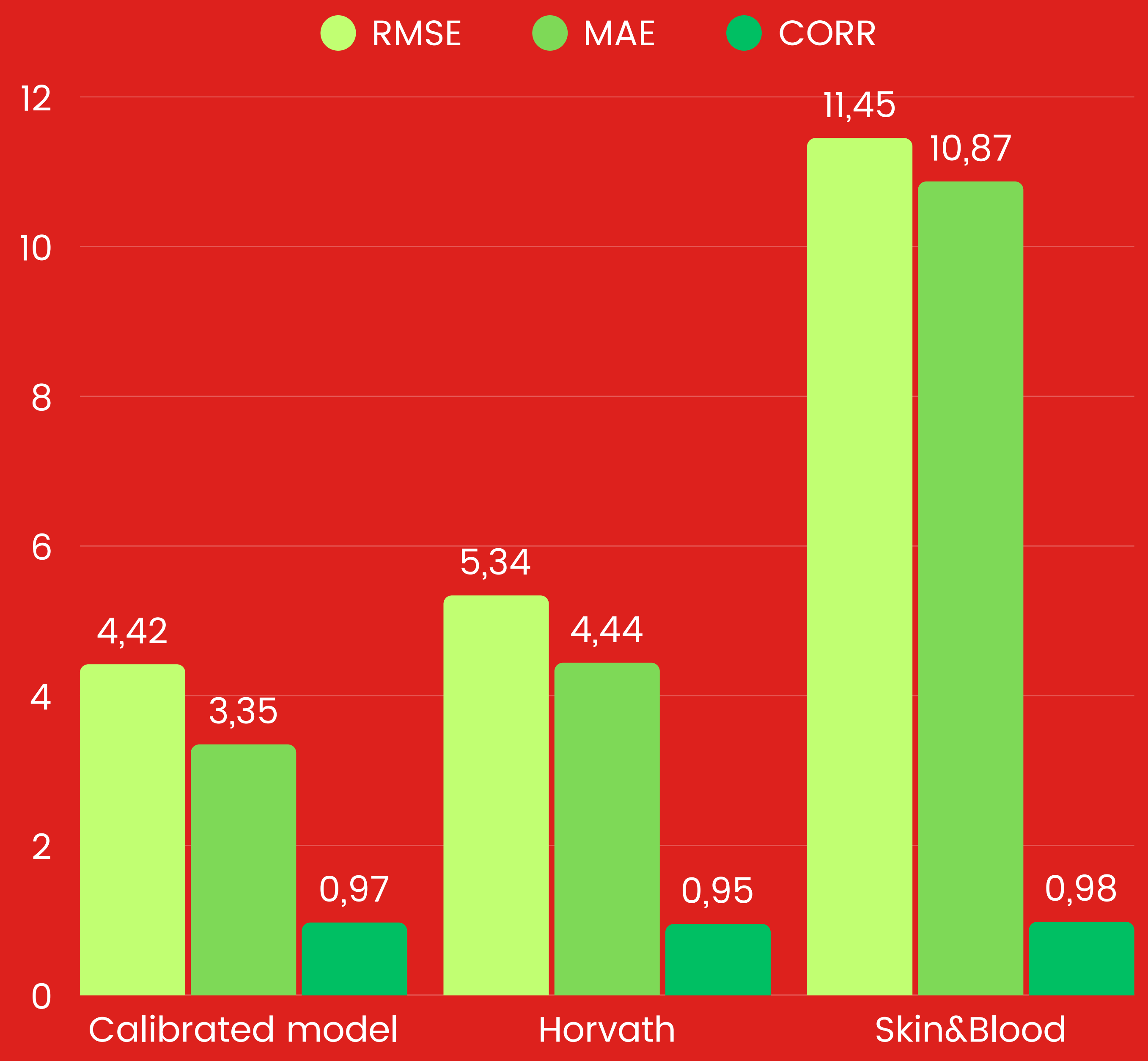


Horvath pantissue model

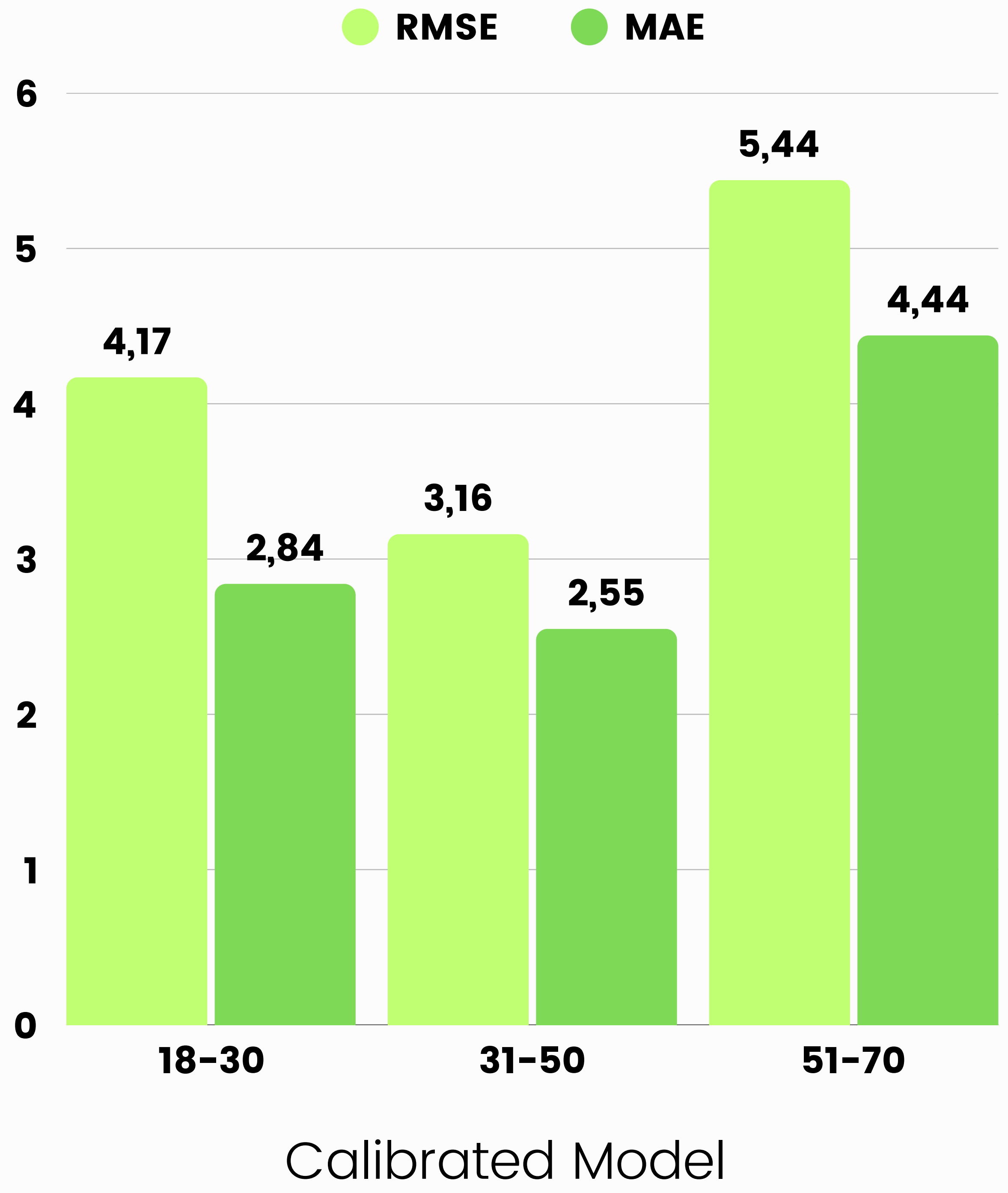


Skin & Blood model

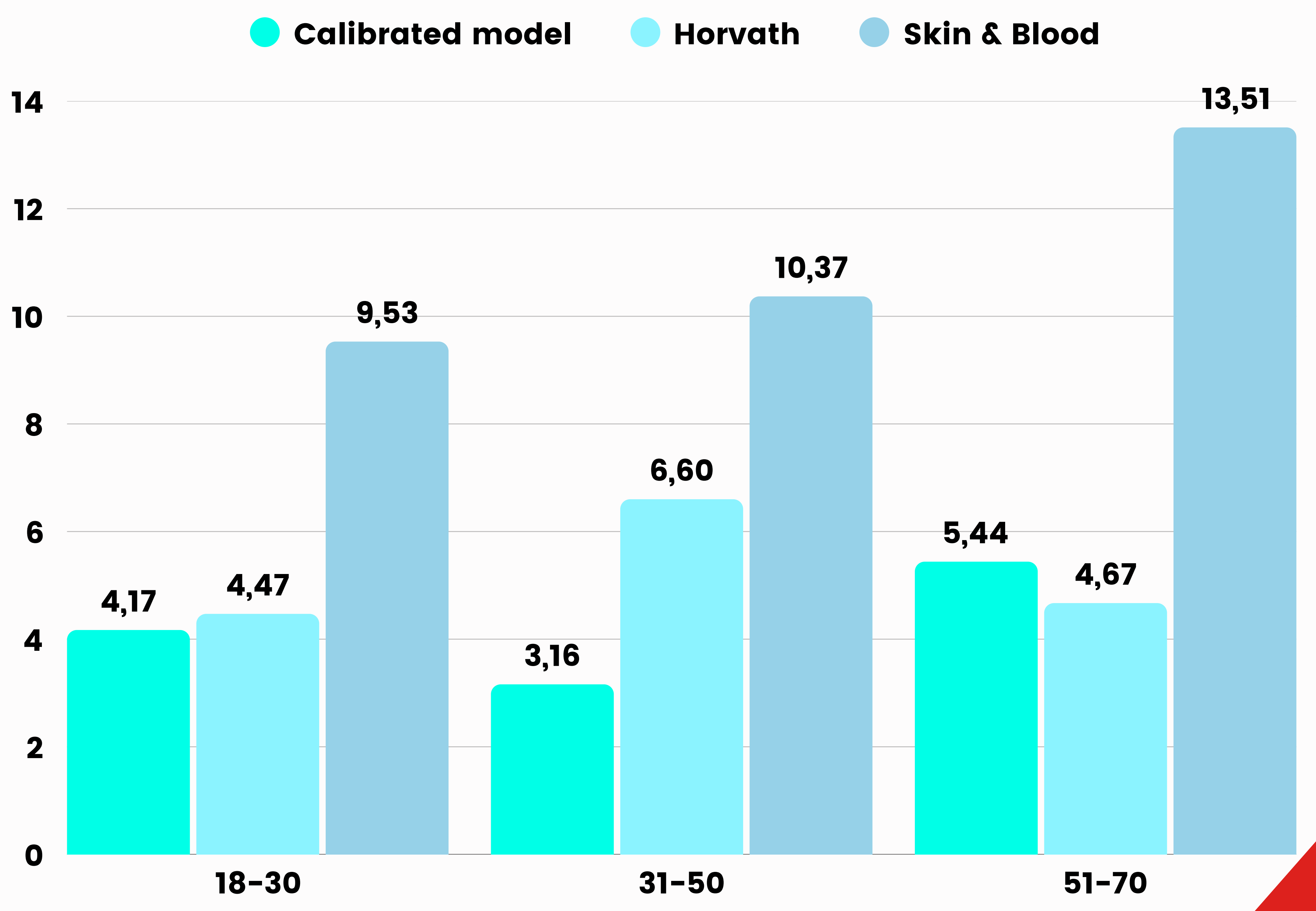
Overall results



Results by age group



RMSE by age group



Results **Analysis**

Covariates impact on predictions

Δage (predicted – chronological age) was used to evaluate the effect of ethnicity and sex on aging acceleration/ deceleration.

Data enrichment

The model's selected probes were first annotated for chromosome location and nearby genes using the IlluminaHumanMethylationEPICv2 package. Those genes were then enriched through Ensembl to gather identifiers, functional details, protein domains and pathway associations.

Biological analysis

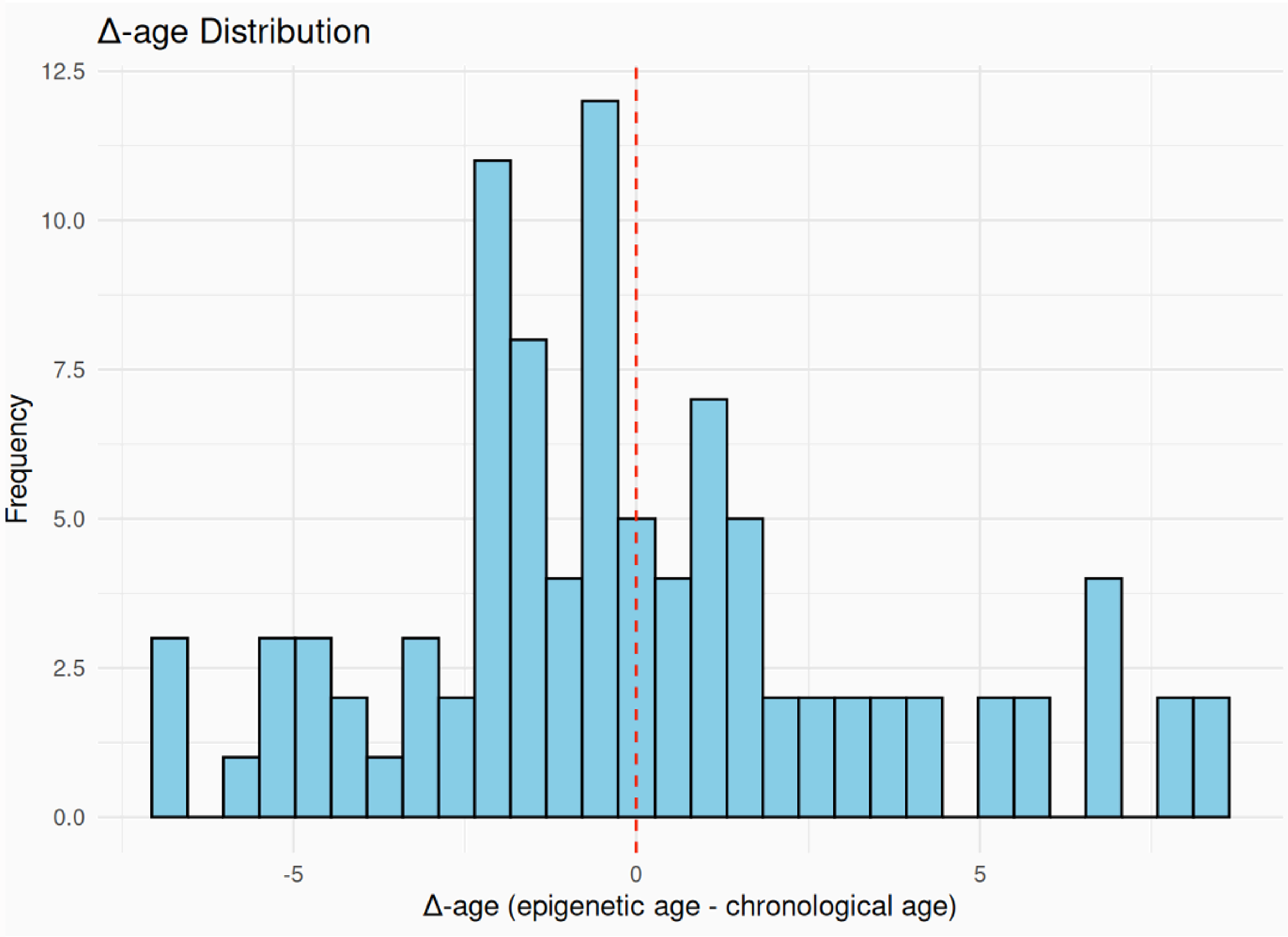
The resulting data were used for exploratory analyses to identify age-predictive genes' molecular functions and associated pathways.

Delta Age

The distribution is centered around zero, indicating good overall calibration

Slight positive skew: some individuals age 5–8 years faster.

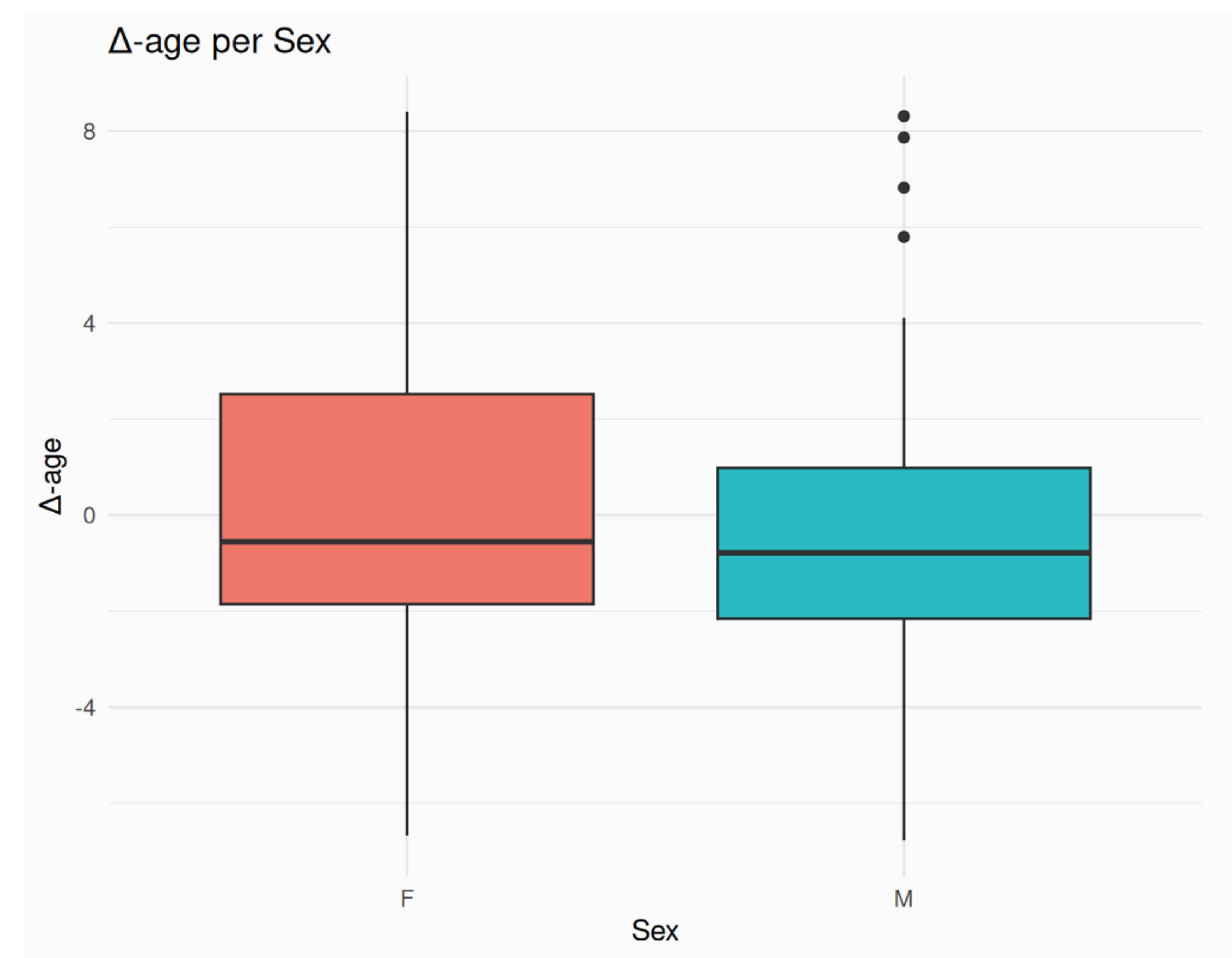
Presence of outliers suggests marked acceleration/deceleration of biological aging.



Delta Age per Sex

Shows sex-based differences in biological aging.

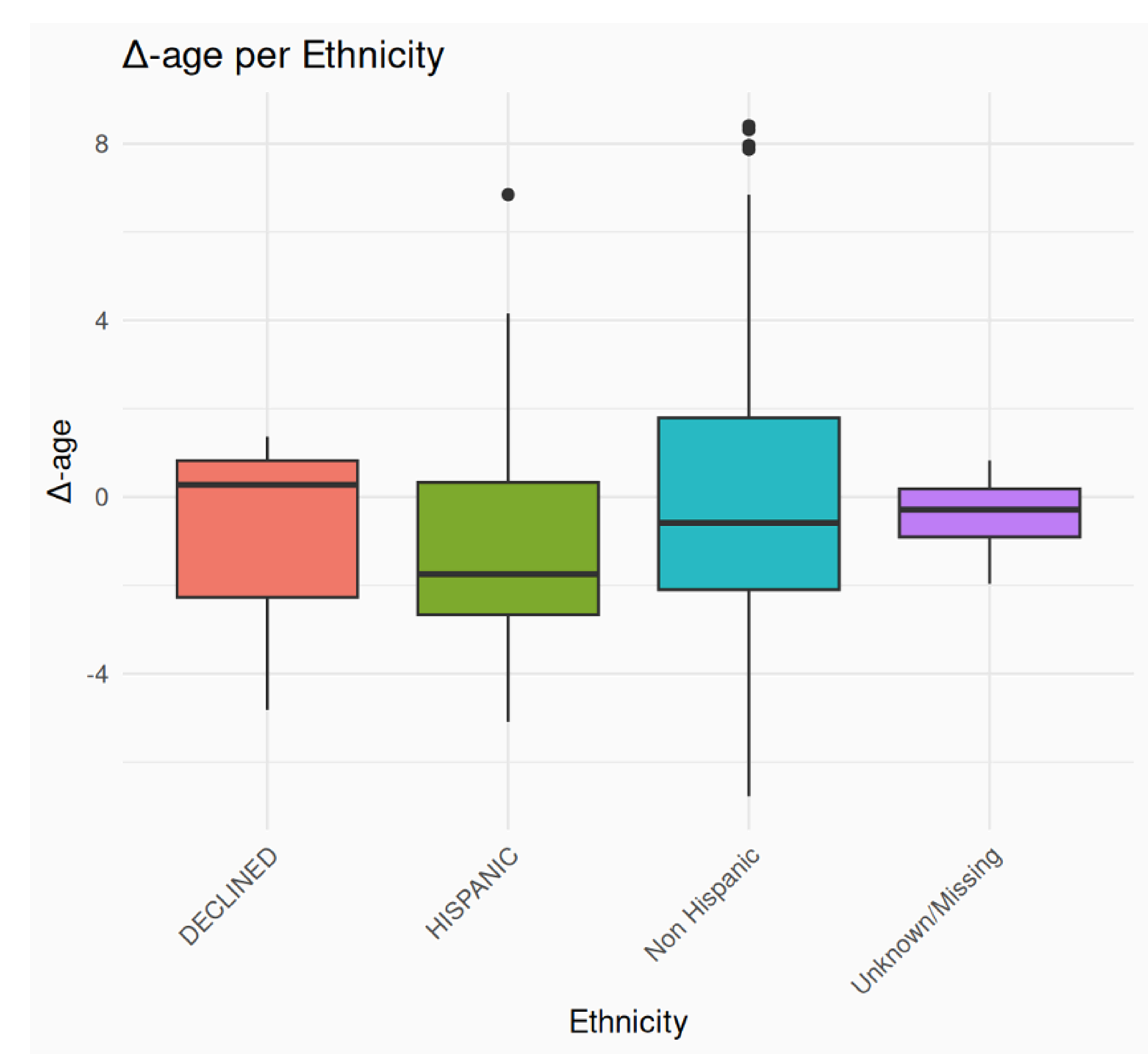
Females exhibit a wider Δ-age range, indicating greater variability in predicted biological age compared to males.



Delta Age per Ethnicity

Examines Δ-age across ethnic groups, primarily: Hispanic and Non-Hispanic.

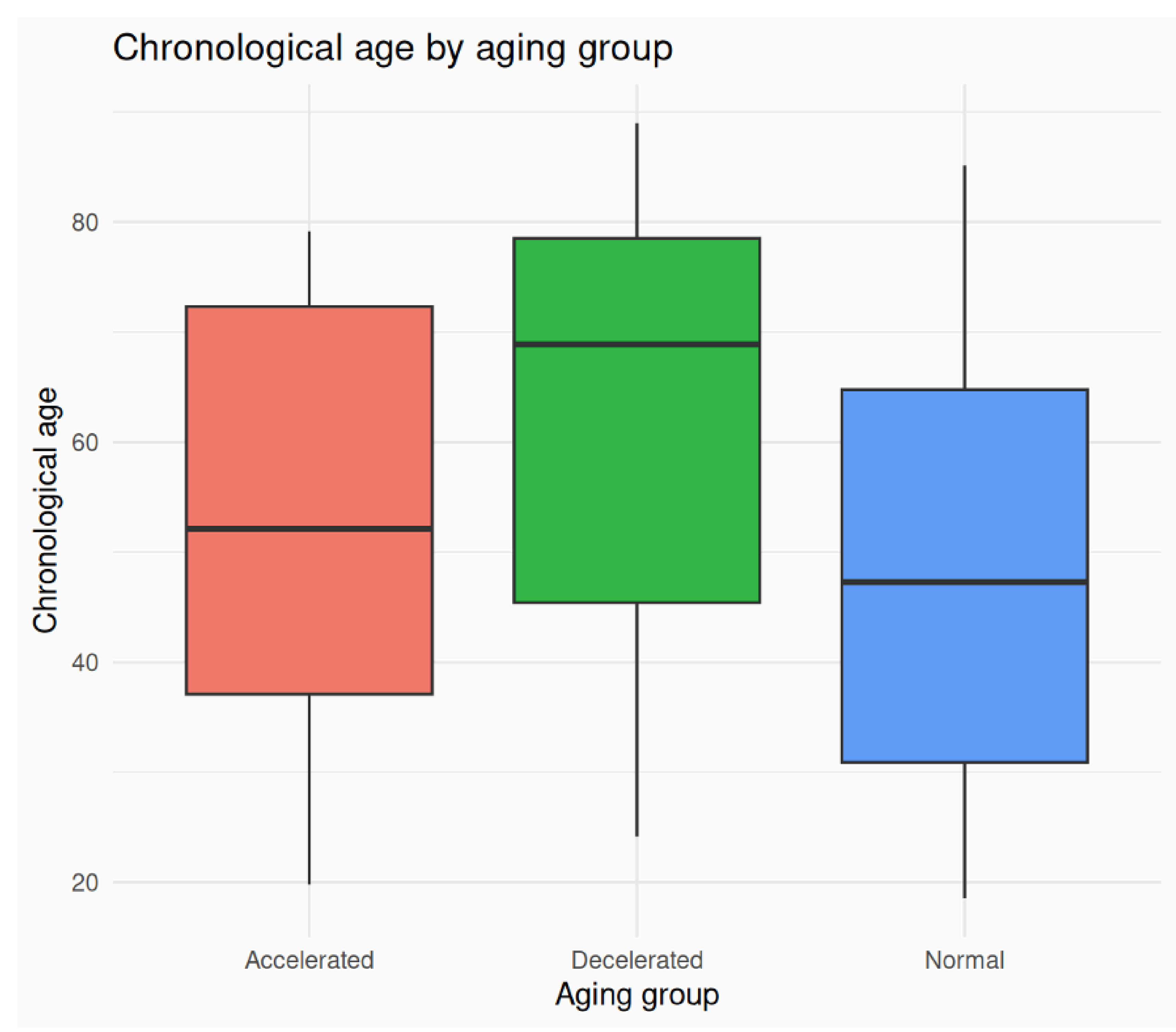
Hispanics tend to have a lower predicted age (mean Δ-age ≈ -2).



Delta Age - aging groups

Compares chronological age across three groups: accelerated (Δ-age > 2), decelerated (Δ-age < -2), and normal.

Highlights the systemic bias: older individuals tend to show deceleration over youger ones.



Data Enrichment

The probes selected by the model were annotated using the IlluminaHumanMethylationEPICv2 package, which provides for each probe:

- chromosome and genomic position
- genes located in the probes targeted region

The identified genes were then further enriched via the Ensembl database, retrieving for each gene:

- HGNC symbol: official symbol of the gene
- gene biotype: functional description of the gene
- InterPro domains: protein domains
- Reactome pathways: associated pathways
- GO terms (name_1006): gene ontology (molecular function, cellular component and biological process)

Age Predictive Genes

The probes selected by the model target a total of 61 genes, primarily located on chromosomes 1, 2, 5, 6 and 17.

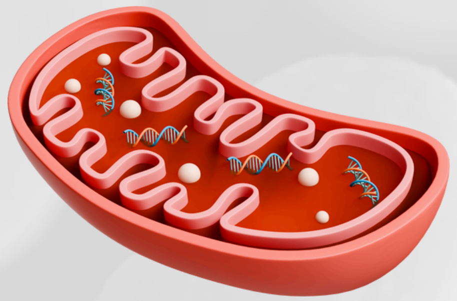
This distribution is expected, as chromosomes 1 and 2 are the largest in the human genome.
These chromosomes are enriched in gene clusters involved in regulatory processes, transcription factors, and immune response pathways.

Genes by probe coverage

SOD2 SCN5A RPA2

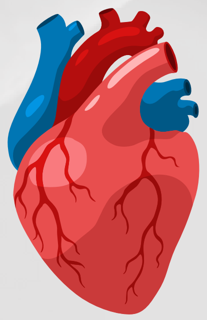
SOD2

A mitochondrial enzyme from the manganese superoxide dismutase family. It converts superoxide radicals into hydrogen peroxide and oxygen, protecting cells from oxidative stress. It's associated with apoptosis, transcription and regulation activities. It's implicated in aging, cardiomyopathy, cancer, and neurodegeneration.



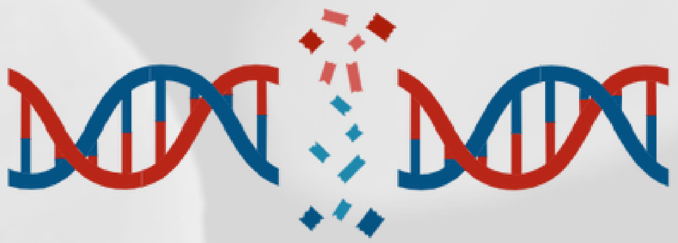
SCN5A

Encodes a cardiac sodium channel essential for initiating action potentials. It's involved in cardiac excitability.



RPA2

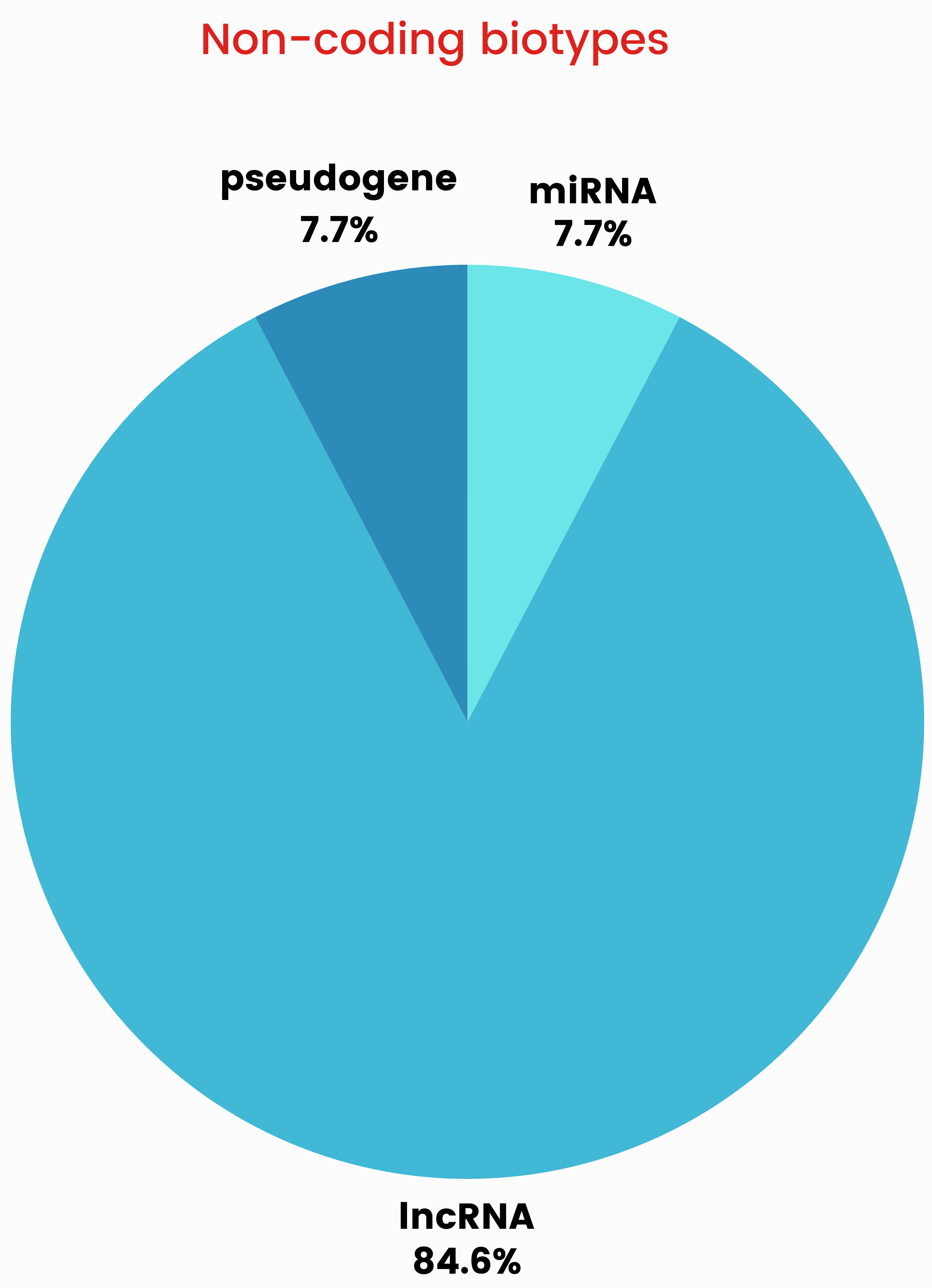
Part of the RPA complex that binds single-stranded DNA during replication and repair. It plays a key role in maintaining genome stability and coordinating DNA damage response.



Gene Biotype

The majority of age predictive genes are protein-coding (48 over 61), indicating a strong focus on genes involved in active protein synthesis.

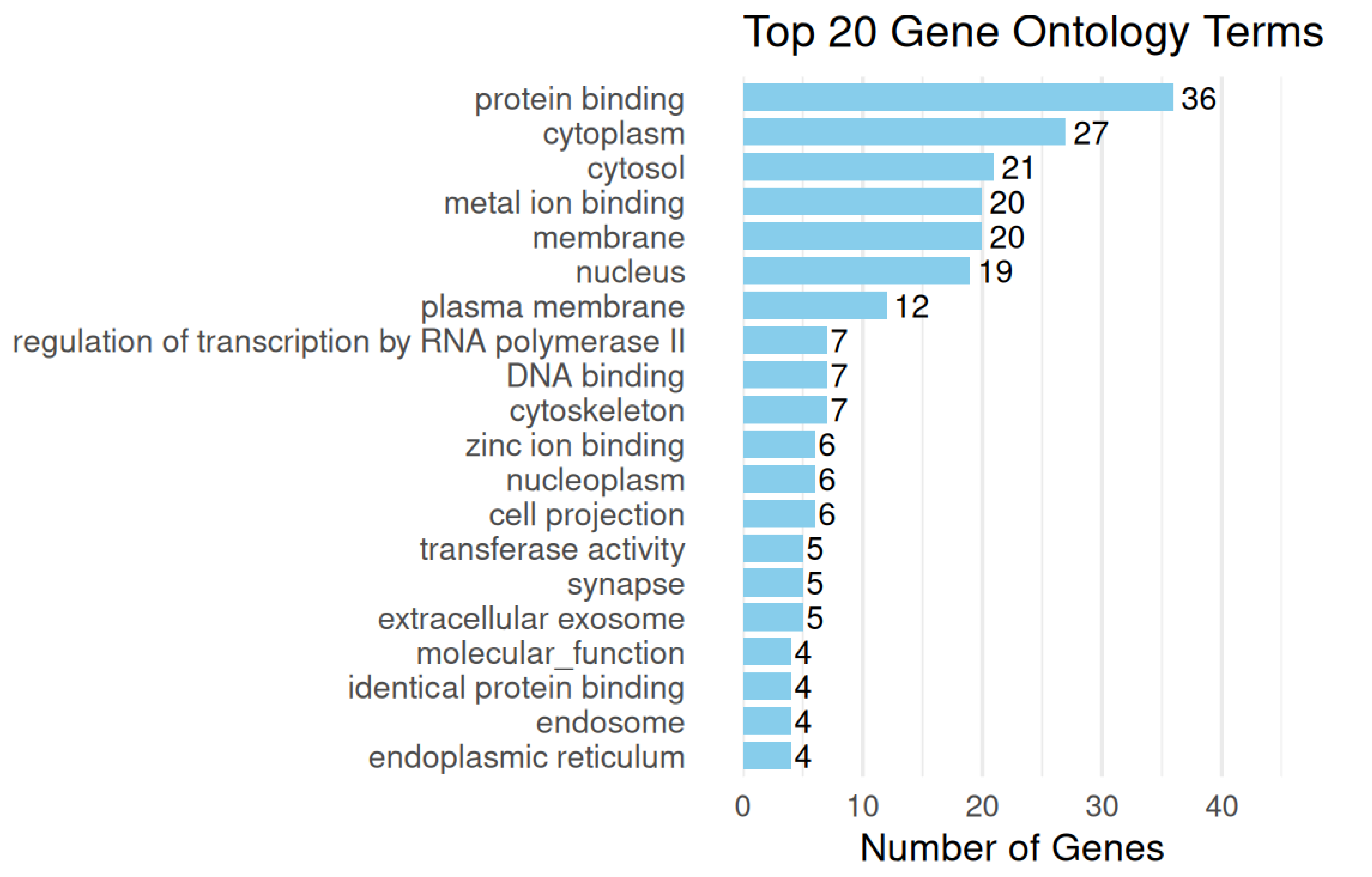
Non-coding biotypes are far less represented (1 pseudogene, 1 miRNA and 11 lncRNA).



Gene Ontology

The Gene Ontology terms highlight the key biological roles of age predictive genes in:

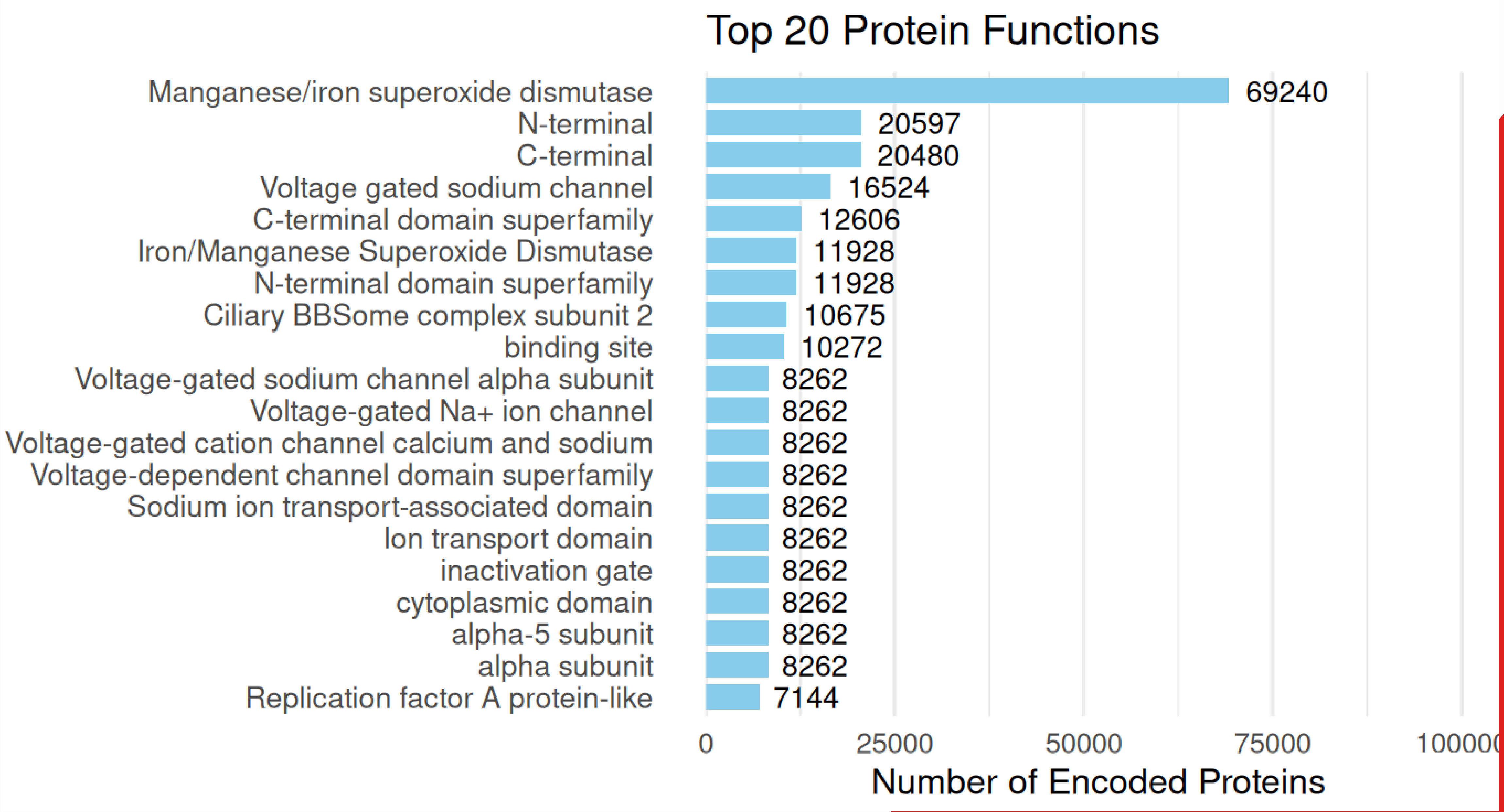
- Enzymatic activity (e.g., oxidative stress response),
- Metal ion signaling (e.g., calcium binding)
- Transcriptional regulation



Protein Functions

The proteins most encoded by age predictive genes perform as:

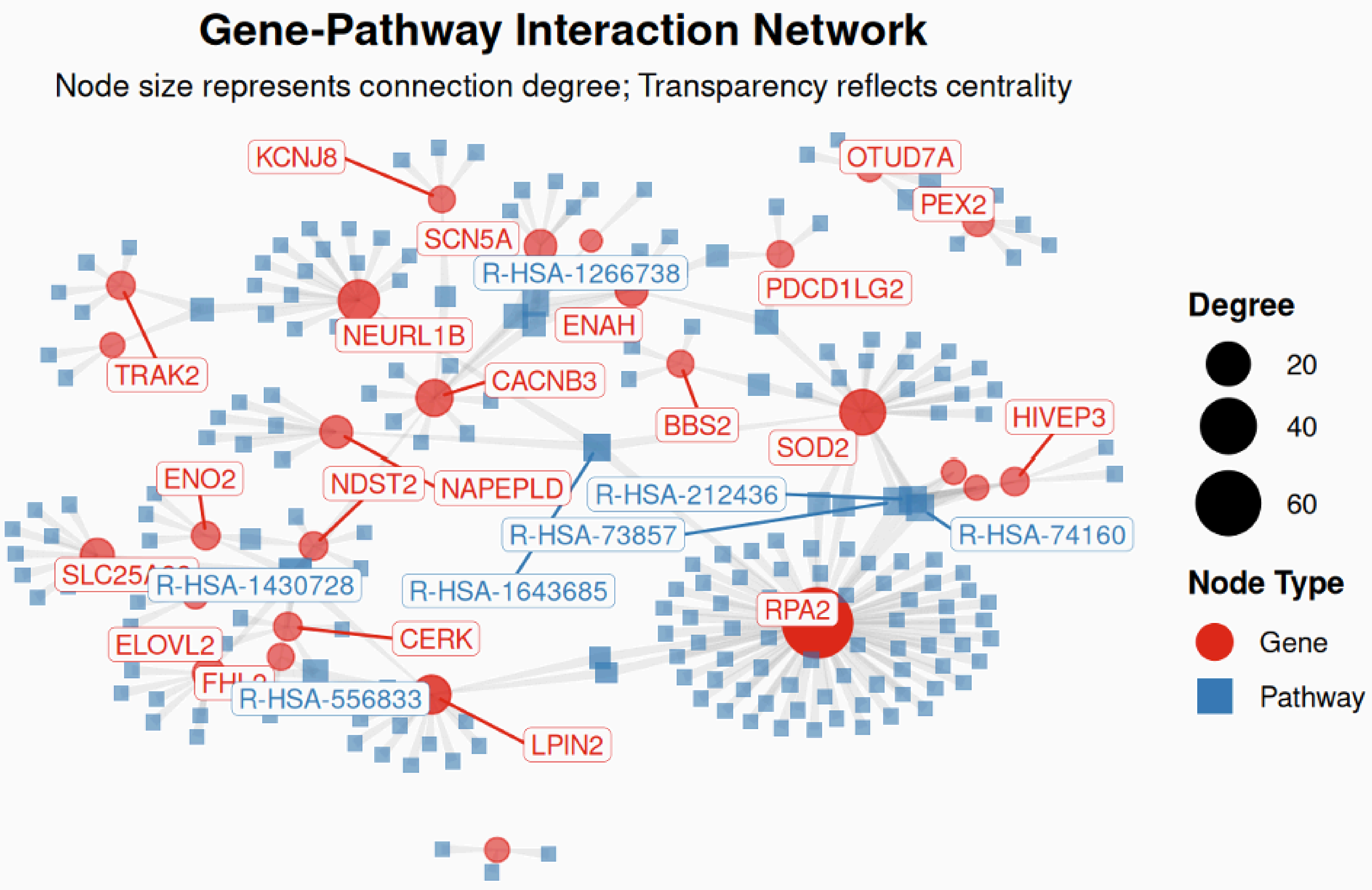
- Superoxide Dismutase: involved in detoxifying reactive oxygen species and protecting cells from oxidative stress.
- Voltage-Gated Ion Channels (e.g., SCN5A, CACNB3): involved in cellular electrical signal transmission, especially in heart and nervous system.
- DNA Replication .



Gene Pathway Network

The network highlights interactions between key hub genes (SOD2, SCN5A, RPA2, LPIN2) and Reactome pathways:

- Developmental Biology (R-HSA-1266738): involved in tissue development and cell differentiation (SCN5A is linked to cardiac development).
- Metabolism (R-HSA-1430728): covers essential metabolic processes such as nutrient cycling.
- Disease (R-HSA-1643685): including metabolic disorders and neurodegenerative conditions.
- Metabolism of lipids (R-HSA-556833): implicated in the process of synthesizing, breaking down, and storing fat in cells for energy storage (LPIN2 regulates lipid metabolism).

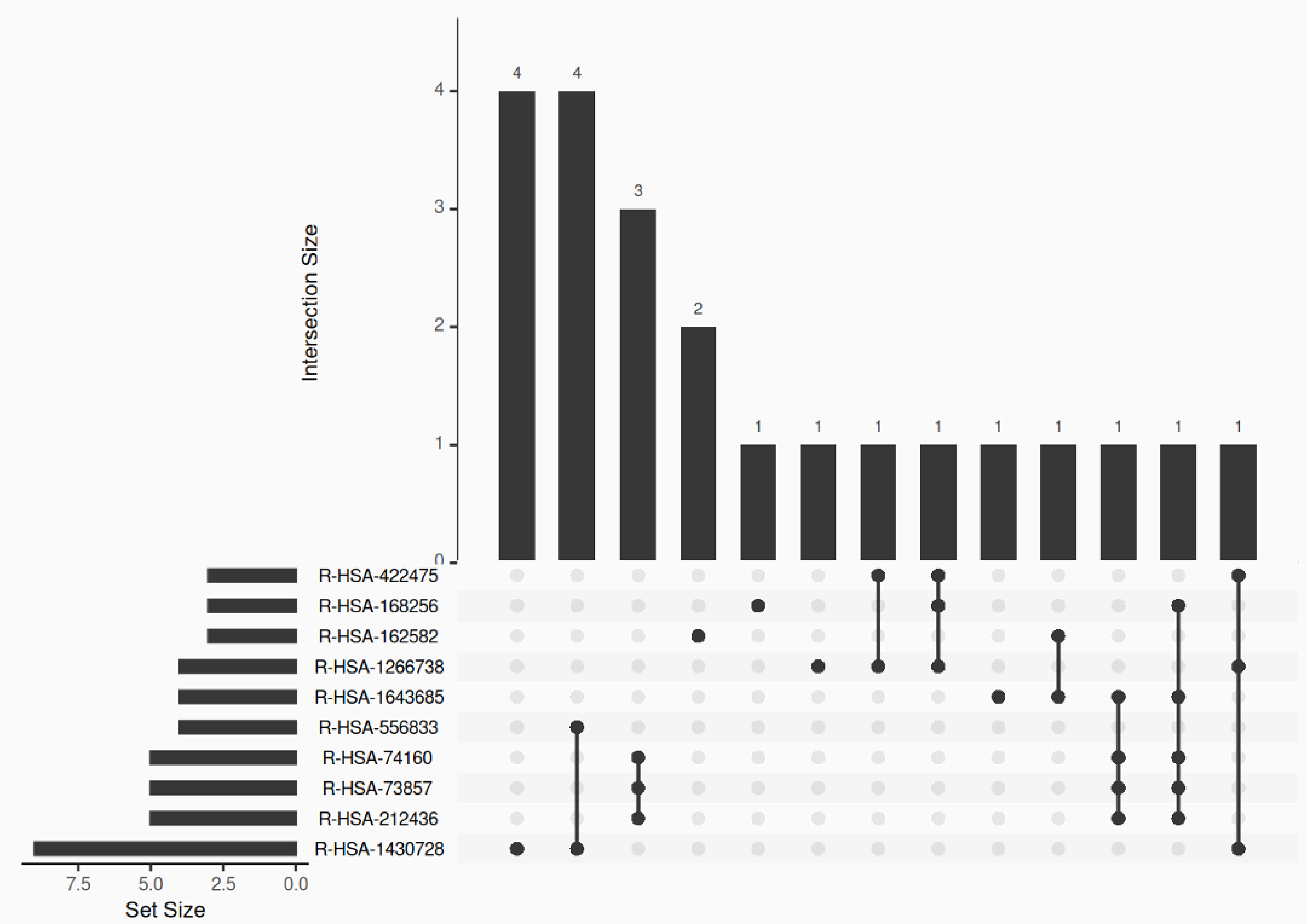


Red circles: Genes | Blue squares: Pathways

Gene Pathway UpSet plot

The UpSet plot provides an alternative to network visualization, highlighting gene-pathway overlaps. It reveals that several genes are multifunctional, participating in multiple biological processes simultaneously.

- Axon Guidance(R-HSA-422475): involved in axonal navigation; relevant in neurological disorders.
- Immune System(R-HSA-168256): enriched in genes associated with inflammatory response.
- Signal Transduction(R-HSA-162582): covers membrane receptor-mediated signaling.

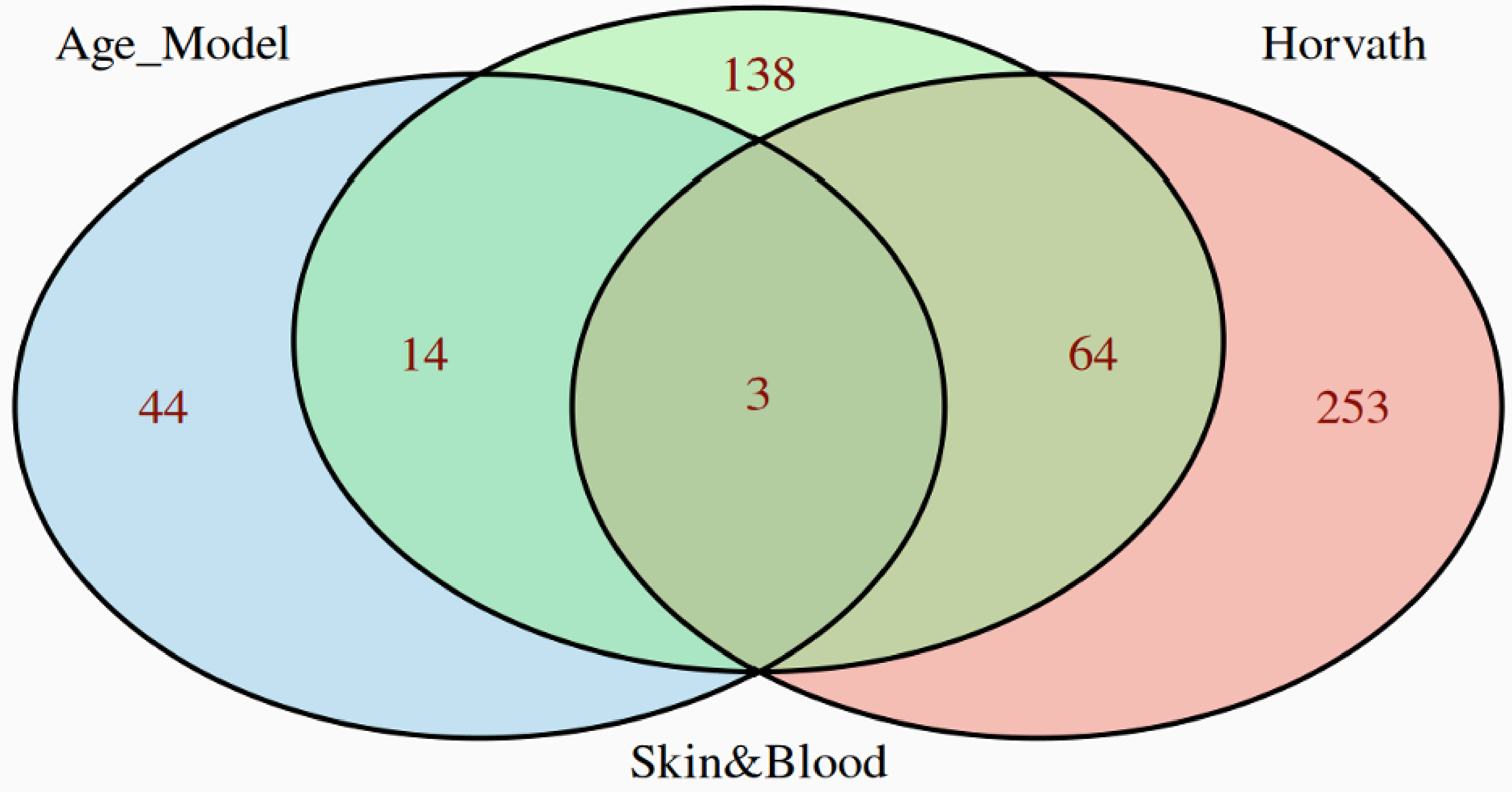


Overlay Age Predictive Genes

An overlap analysis was performed between genes selected by:

- The calibrated age model,
- Horvath's pan-tissue model,
- And the Skin & Blood model.

Three genes were shared across all three models:
KLF14, PRR34-AS1, and SCGN



KLF14 PRR34-AS1 SCGN

KLF14
A Krüppel-like zinc-finger transcription factor located in the nucleus. It plays a key role in lipid metabolism and gene expression. Also acts as a TGF-β–responsive co-repressor and is maternally expressed (it's linked to metabolic diseases like diabetes).

PRR34-AS1
A long non-coding antisense RNA (lncRNA), involved in transcriptional regulation of the sense gene PRR34. It's associated with glioma susceptibility and childhood medulloblastoma.

SCGN (Secretagogin)
Encodes a calcium-binding protein involved in synaptic function and intracellular calcium regulation, particularly in neurons. It's related to insulinoma and neuroendocrine tumors.

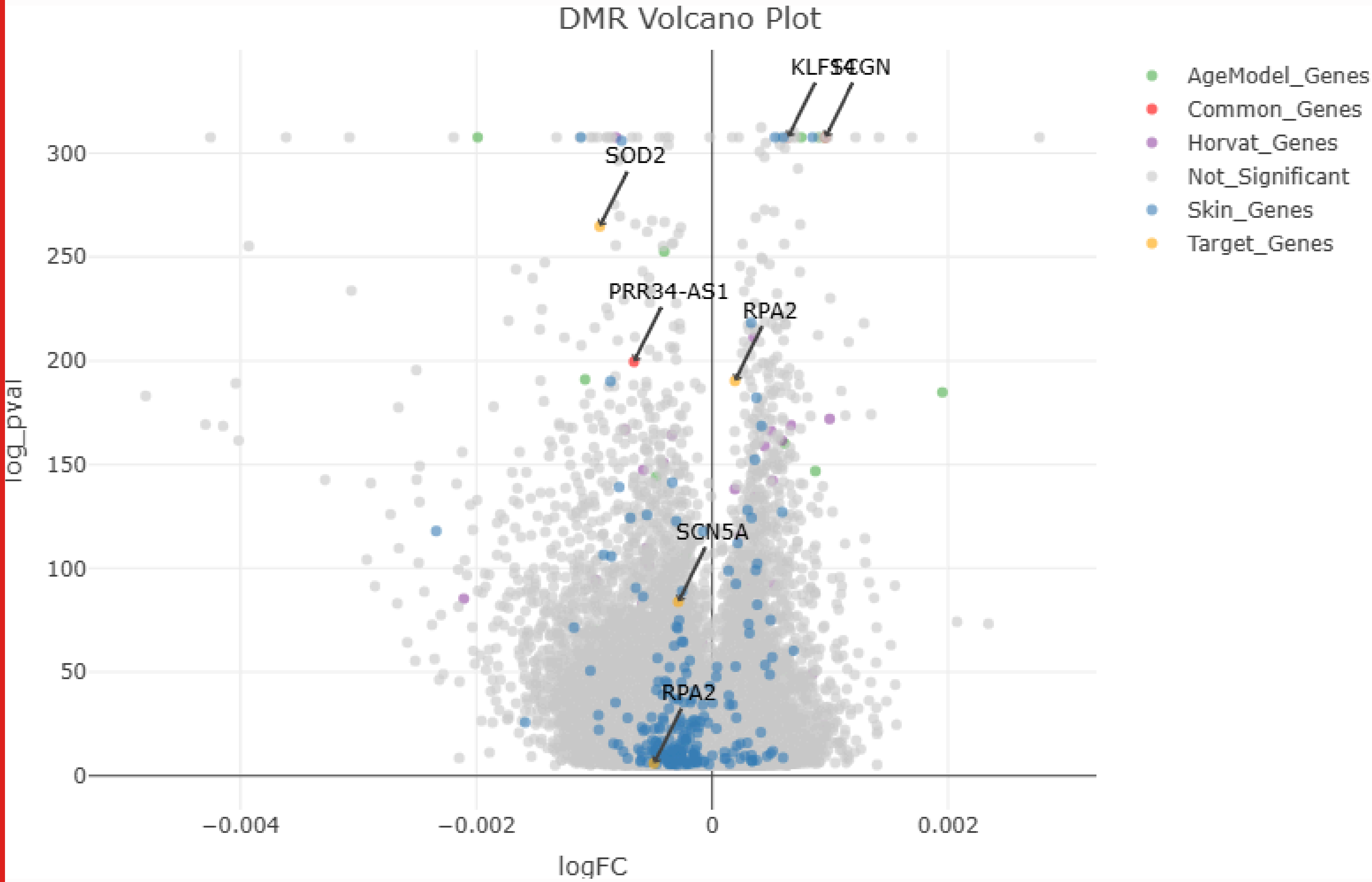
The TGF-β protein family is part of the transforming growth factor-beta superfamily.
It plays a central role in regulating cell proliferation, differentiation, and other key functions in most cell types.

Age Associated Methylation Changes

Differentially methylated regions identified with **DMRcate** package on all available data, correlating methylation changes with age.

Only a few genes exhibit significant age-related methylation changes:

- PRR34-AS1,SOD2, SCN5A, RPA2: hypomethylated in older individuals, potential aging biomarkers.
- KLF14, SCGN, RPA2: hypermethylated, suggesting a possible reduction in their gene expression in older individuals.



Project Conclusions

Key Findings:
The predominant demethylation of age predictive genes with increasing age suggests likely upregulation.

The hypermethylation observed in a subset of genes remains uncertain whether it reduces expression or reflects broad, non-specific regional methylation.

Impacted functions:

- Oxidative stress protection (SOD2)
- Cardiac activity (SCN5A)
- DNA repair (RPA2)
- Calcium binding (SCGN)
- Lipid metabolism (KLF14)

These pathways align with age-related pathologies (cardiovascular, neurodegenerative, cancer).